

9 (long)

(a) X and Y are two 2-bit unsigned binary numbers, where $Y > 1$. The 4-bit binary number P satisfies the equation

$$P = (X)^2 + Y$$

Design a digital circuit that evaluates the most significant bit and the least significant bit of P from the bits of X and Y . The circuit should be composed only of NOR gates and the smallest possible number of such gates should be used.

Explain why designs with larger numbers of gates can be beneficial. [15]

(b) 2-bit binary unsigned numbers are received sequentially by a detector. Each of these numbers is greater than 1. The output Z of the detector becomes 1 if the current value of the number received is greater than 2 and the value of the previous number received was equal to 2. The output Z is otherwise 0.

(i) Draw a state diagram for the system. [6]

(ii) State how many JK bistables will be needed to implement the detector and draw a state transition table (assume that unused states are never reached). [5]

(iii) Derive simplified Boolean expressions for the J inputs of the bistables. [4]